

Mobile Application Development

Continuous Assessment Project

SA62

Project

AI-Enabled Wellness Mobile App

Assignment Overview

In this project, your team will build a simple AI-enabled wellness mobile application.

Mandatory Features

Your app should allow users to:

- support user login and logout in the mobile application
- enter and view basic wellness records (example of wellness records - sleep hours, exercise activities and duration etc.)
- connect to a backend service
- store and retrieve data from a MySQL database (through backend services)
- interact with a simple chatbot

This project is designed to help you demonstrate your understanding of:

- mobile app development
- backend integration (e.g. with RESTful services)
- database persistence
- prompt engineering or RAG
- end-to-end system integration

Team Project

You should complete this project as a team

Deliverables

Your solution must include the following components:

1. Android Mobile App

Build a mobile app using Kotlin. The app should provide a user-friendly interface for:

- entering wellness-related data
- viewing previously saved records
- chatting with the AI assistant

2. Java Spring Boot Backend

Build a backend service using Java Spring Boot (or .Net). The backend should:

- provide APIs for the mobile app
- handle business logic
- connect to the database
- send chatbot requests to the AI service

3. MySQL Database

Use MySQL to store application data such as:

- user information
- wellness logs
- chatbot responses or recommendations

4. Simple Chatbot

Your app must include a chatbot that demonstrates prompt engineering or RAG.

The chatbot should:

- accept user questions from the mobile app
- generate relevant wellness-related responses
- work through the backend

Authentication:

- Login with JWT authentication
- Python-based agentic AI feature

Agentic AI Feature:

Teams may implement a Python-based agentic AI component to demonstrate a more advanced AI workflow. Unlike a simple AI-enabled chatbot that only responds to user

prompts, an agentic AI feature should show some degree of autonomy, such as retrieving user data, analysing trends, making simple decisions, and generating or saving recommendations. The agent may be triggered on demand by the user or automatically by the backend at scheduled intervals (e.g. via a Spring Boot scheduler).

A simple agent workflow could be as follows:

1. retrieve a user's recent wellness records
2. review the user's recent trends
3. generate a personalised wellness recommendation
4. save the recommendation
5. display it in the mobile app

The agent may be triggered:

- when the user requests it, or
- automatically by the backend using a scheduled task (e.g. via a Spring Boot scheduler)

Scope

The number of features should be at least as many as the number of team members. The features should be **integrated** as one digital solution.

Features	Number of Features	Remarks
Login + Logout	1	Mandatory
Create / Update / Delete / Retrieve current or historical wellness data	Each counted as 1 feature	Mandatory
Chatbot	1	Mandatory
Agentic AI feature	1	
Securing the apps with JWT	1	

Submission

1. Please work in your assigned team for the project.
2. The **deliverable** for this CA is a **single zipped file_** that consists of:
 - a. One Integrated Solution with the codes.
 - b. Video Demo (try to keep it to 15 mins)
 - Demonstrating how users use your system.

3. Indicate the author in all the classes or methods. It is possible to have more than one author for the same class or methods.
4. Each team should only have ONE submission. Please name your submission with your team's name. For example, **if you are Team 1**, then your submission would be **Team1.zip**.
5. The submission deadline: **9th Jul (Thursday), 11pm.** Please upload your deliverable to the **CA Submission Assignment folder** in Canvas by the deadline.

Presentation Schedule

Date: 10 Jul (Fri)

Time	Accessor 1 Venue: TBC
9:40 – 10:00	Team 1
10:00 – 10:20	Team 2
10:20 – 10:40	Team 3
10:40 – 11:00	Team 5
11:00 – 11:20	Team 6
14:00 – 14:20	Team 7
14:20 – 14:40	Team 8
14:40 – 15:00	Team 9

Assessment Criteria

- **Requirements Fulfilment**
How well the application meets the stated project requirements, including mobile app, backend integration, database persistence, and chatbot implementation.
- **Functionality and Integration**
Whether the application works as expected, with smooth interaction across the mobile frontend, Java Spring Boot backend, MySQL database, and AI/chatbot components.
- **Code Quality**
The extent to which the code is readable, maintainable, well-structured, and follows good development practices.
- **User Interface and User Experience**
How intuitive, clear, and user-friendly the mobile application is.
- **System Design**
The quality of the overall solution design, including architecture, API design, and component integration.
- **Bonus / Advanced Features**
Additional features such as secure communication (JWT) or optional agentic AI capabilities.

Marking

Breakdown:

	Items	Marks
1.	Main Features	35
2.	Peer Evaluation	5
	Total	40